SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

SIMATS ENGINEERING

Assessment Tool 3 — Comparative Analysis

Course Name: Computer Networks | Course Code: [illegible]

Course Outcome: CO4 — Design network topologies (star, mesh, bus, hybrid) using networking devices, transmission media, and cabling standards

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Q1. Compare Star topology and Bus topology.

Answer: Star topology connects every node individually to a central device, such as a switch or hub, whereas bus topology connects all nodes along a single shared backbone cable. This structural difference drives most of their trade-offs. In terms of cost, star topology is generally more expensive because it needs a central device plus a separate cable run for every node, while bus topology needs only one backbone cable and minimal connectors, making it cheaper and quicker to install. In terms of reliability, star topology is superior: the failure of one node or its cable is isolated and does not affect the rest of the network, whereas a break anywhere in the bus backbone brings the entire network down. In terms of performance, star topology also has the edge, since the central device manages and directs traffic, reducing the chance of collisions, while bus topology shares one medium among all devices, so performance degrades noticeably as more nodes are added and network traffic increases. Overall, bus topology suits small, low-budget networks where simplicity matters most, while star topology is preferred wherever reliability and consistent performance are priorities, such as in offices and campus LANs.

Q2. Compare Mesh topology and Star topology in terms of fault tolerance.

Answer: Fault tolerance refers to a network's ability to continue functioning when a link or device fails, and this is where mesh and star topologies differ most sharply. In mesh topology, every node is directly connected to multiple other nodes, so several alternate paths exist between any two devices; if one link or even several links fail, data can simply be rerouted through another path, and communication continues uninterrupted. This makes mesh topology extremely fault-tolerant and well suited to critical systems such as military networks, backbone infrastructure, and mission-critical data centers, where downtime is unacceptable. Star topology, by contrast, offers only moderate fault tolerance: because every node connects solely to the central hub or switch, the failure of an individual node or its cable is isolated and harmless to the rest of the network, but the central device itself is a single point of failure — if it goes down, the entire network stops functioning, since there is no alternate path around it. Consequently, mesh topology is the more robust and reliable choice for critical environments, while star topology's fault tolerance depends heavily on the reliability of the central device, which can be improved with redundant or backup switches but never fully matches the inherent path redundancy of mesh.

Q3. Compare Bus topology and Mesh topology in terms of installation complexity.

Answer: Installation complexity depends on how many physical connections a topology requires, and bus and mesh sit at opposite ends of this spectrum. Bus topology is simple and quick to install because it uses one continuous backbone cable, with each device connected to it through a single tap or connector; wiring is minimal, so it is well suited to small networks with a handful of devices. Mesh topology, in contrast, requires every node to be directly connected to every other node, meaning the number of links grows rapidly with the number of devices — for n nodes, a full mesh requires $n(n-1)/2$ individual connections. This demands a large number of cables, network interface ports, and careful planning, making installation time-consuming, labor-intensive, and expensive, especially as the network scales. As a result, bus topology remains attractive for small, cost-sensitive setups, while mesh topology, despite its installation complexity, is chosen where the resulting reliability and redundancy justify the extra cost and effort.

Q4. Compare Hybrid topology and Star topology in terms of scalability.

Answer: Scalability refers to how easily a network can grow to accommodate more devices or segments without major redesign, and hybrid topology has a clear advantage over star topology in this respect. Hybrid topology combines two or more basic topologies, such as star, mesh, or bus, allowing an organization to mix and match structures to fit different parts of the network: new segments, buildings, or departments can be added as additional sub-networks without disturbing the existing structure, which makes hybrid networks highly flexible and scalable, and is why large enterprises and campus networks commonly adopt hybrid designs. Star topology is also reasonably scalable, since new nodes can be added simply by running a cable to the central device, but this scalability is bounded by the physical port capacity of that switch or hub; once the central device reaches its port limit, further expansion requires purchasing and integrating additional hardware, which can introduce extra cost and complexity. Therefore, while star topology scales well for moderate growth within a single central device's capacity, hybrid topology provides substantially greater long-term scalability for large or evolving networks.

Q5. Compare Star, Mesh, Bus, and Hybrid topologies in terms of maintenance.

Answer: Maintenance effort varies considerably across the four topologies because it depends directly on how nodes are interconnected. Star topology is the easiest to maintain: because every device has its own dedicated link to the central hub or switch, faults can be quickly identified and isolated to a specific node or cable without disrupting the rest of the network. Bus topology is comparatively harder to maintain than star, since a fault anywhere along the shared backbone cable can disrupt the entire network, and locating the exact point of failure on a long cable run can be difficult and time-consuming. Mesh topology, despite being highly reliable, is the most difficult to maintain due to the sheer number of interconnections between nodes; troubleshooting requires tracing faults across a dense web of links, which becomes increasingly complex as the network grows.